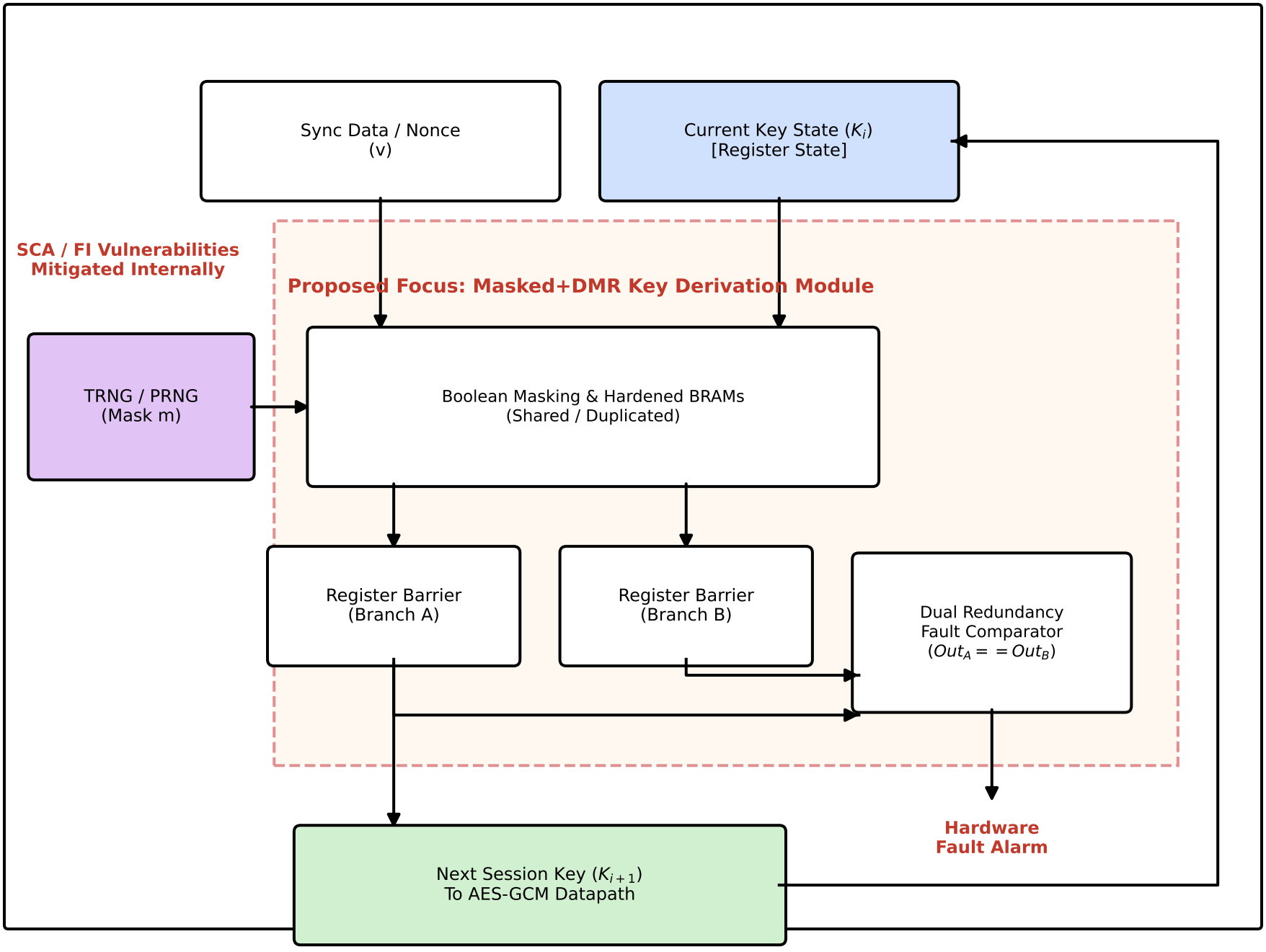
Sync Data / Nonce
(v)
Current Key State (K_i)
[Register State]
SCA / FI Vulnerabilities
Mitigated Internally
Proposed Focus: Masked+DMR Key Derivation Module
TRNG / PRNG
(Mask m)
Boolean Masking & Hardened BRAMs
(Shared / Duplicated)
Register Barrier
(Branch A)
Register Barrier
(Branch B)
Dual Redundancy
Fault Comparator
($Out_A == Out_B$)
Hardware
Fault Alarm
Next Session Key (K_{i+1})
To AES-GCM Datapath